MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE LDP

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NAND gate module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.22 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3 4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be LDP.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/9-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

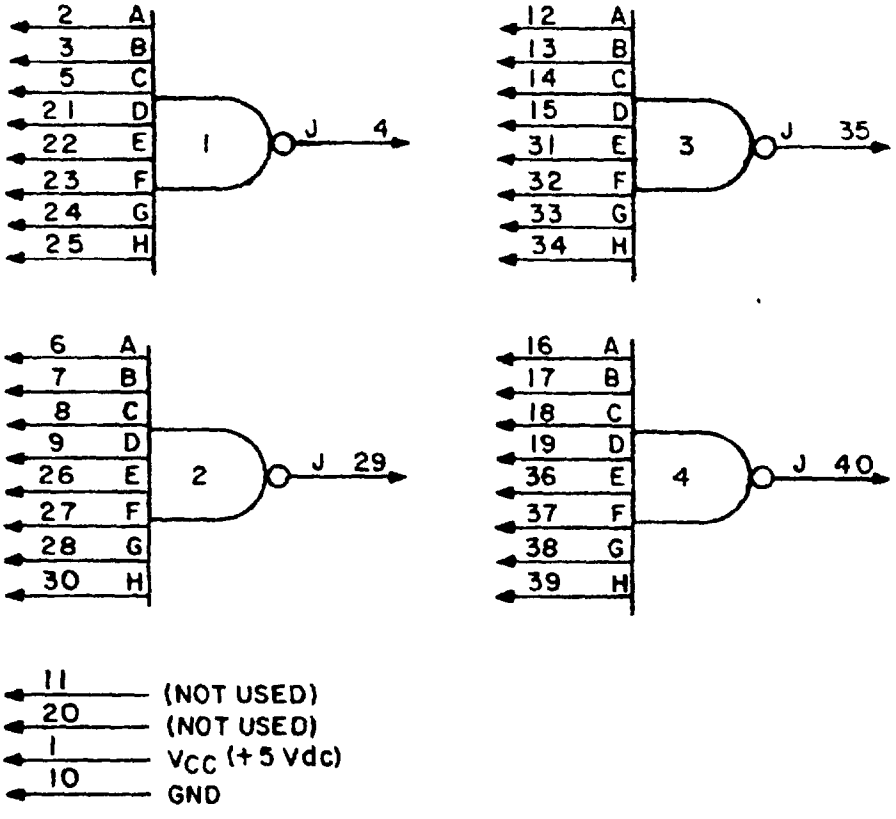
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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* FIGURE 1. LDP module function diagram, NAND gate.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A4	V_{OUT} (low-loaded)	III	S	10	800	10	800	10	800	10	800	mV
B1-B32	Input breakdown current	IV	S	0	1	0	1	0	1	0	1	mA
C1-C4	Output voltage, low level	V	X	0	400	0	400	0	400	0	400	mV
D1-D32	Output voltage, high level	VI	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
E1-E32	Input current, low level	VII	X	0.40	1.73	0.40	1.73	0.40	1.73	0.40	1.73	mA
F1-F32	Input current, high level	VIII	X	0	30	0	30	0	40	0	40	µA
G1-G4	Output short circuit current	IX	X	20	55	20	55	20	55	20	55	mA
H1	Power supply current, low level	X	S	1	24	1	24	1	24	1	24	mA
H2	Power supply current, high level	X	S	1	8	1	8	1	8	1	8	mA
J1-J4	Transition time (t_{TLH})	XI	S	4	45	4	45	4	50	4	50	ns
K1-K4	Transition time (t_{THL})	XI	S	4	20	4	20	4	25	4	25	ns
L1-L4	Propagation delay time (t_{PHL})	XII	S	4	25	4	25	4	27	4	27	ns
M1-M4	Propagation delay time (t_{PLH})	XII	S	4	28	4	28	4	30	4	30	ns
N1-N4	Output crosstalk, low level	XIII	Q	0	400	0	400	0	400	0	400	mV
P1-P4	Output crosstalk, high level	XIV	Q	0	400	0	400	0	400	0	400	mV
R	Power supply transient amplitude	XV	Q	0	400	0	400	0	400	0	400	mV
S	Power supply transient rise time	XVI	S	1	2	1	2	1	2	1	2	ms

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
T1	Isolation	XVII	S	10	-	1	-	10	-	1	-	MΩ
T2-T3	Unused pin isolation	XVII	Q	10	-	1	-	10	-	1	-	MΩ
U1-U32	Input capacitance	XVIII	Q	2	15	2	15	2	15	2	15	pF
V1-V32	Input clamp diode voltage	XIX	Q	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through V shall be performed in any order with the following exceptions. If test conditions A, B, or T are part of a sequence of tests, the order of testing shall be as follows: A, B, T where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through V in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through M and S shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition S, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through T and V shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through V shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Military Specification MIL-M-28787/9 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2833410. Part number M28787/9-1 supersedes part numbers 2833410 and M28787/9.

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	132 mW (low level) 44 mW (high level)
Input voltage	
Low level	0.8 V maximum
High level	2.0 V minimum
Input current	
Low level (V_{IL} = 0 V)	1.73 mA maximum
High level (V_{IH} = 2.4 V)	40 µA maximum

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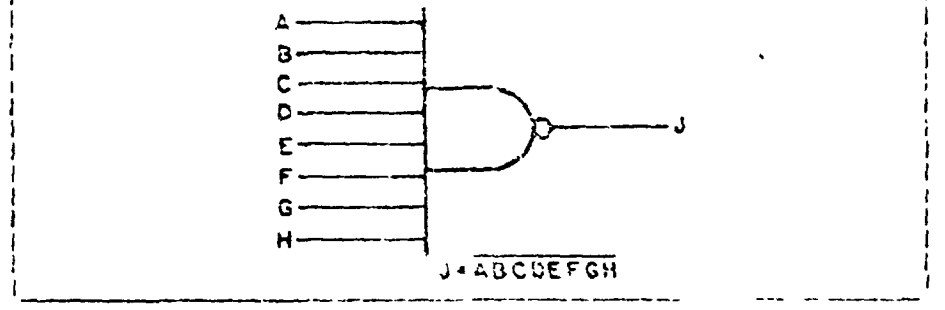
Output voltage	
Low level	0.4 V maximum
High level	2.4 V minimum, VCC maximum
Output current	
Low level (V_{OL} = 0.4 V)	16.0 mA maximum
High level (V_{OH} = 2.4 V)	400 µA minimum
Propagation delay time	
t_{PHL}	4 ns minimum, 27 ns maximum
t_{PLH}	4 ns minimum, 30 ns maximum

* 6.5 Module function. This module contains four independent 8-input NAND gates as illustrated on figure 1.

6.5.1 Input to output relationship. Each NAND gate has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship

Input	Output
At least one input = L	H
All inputs = H	L



6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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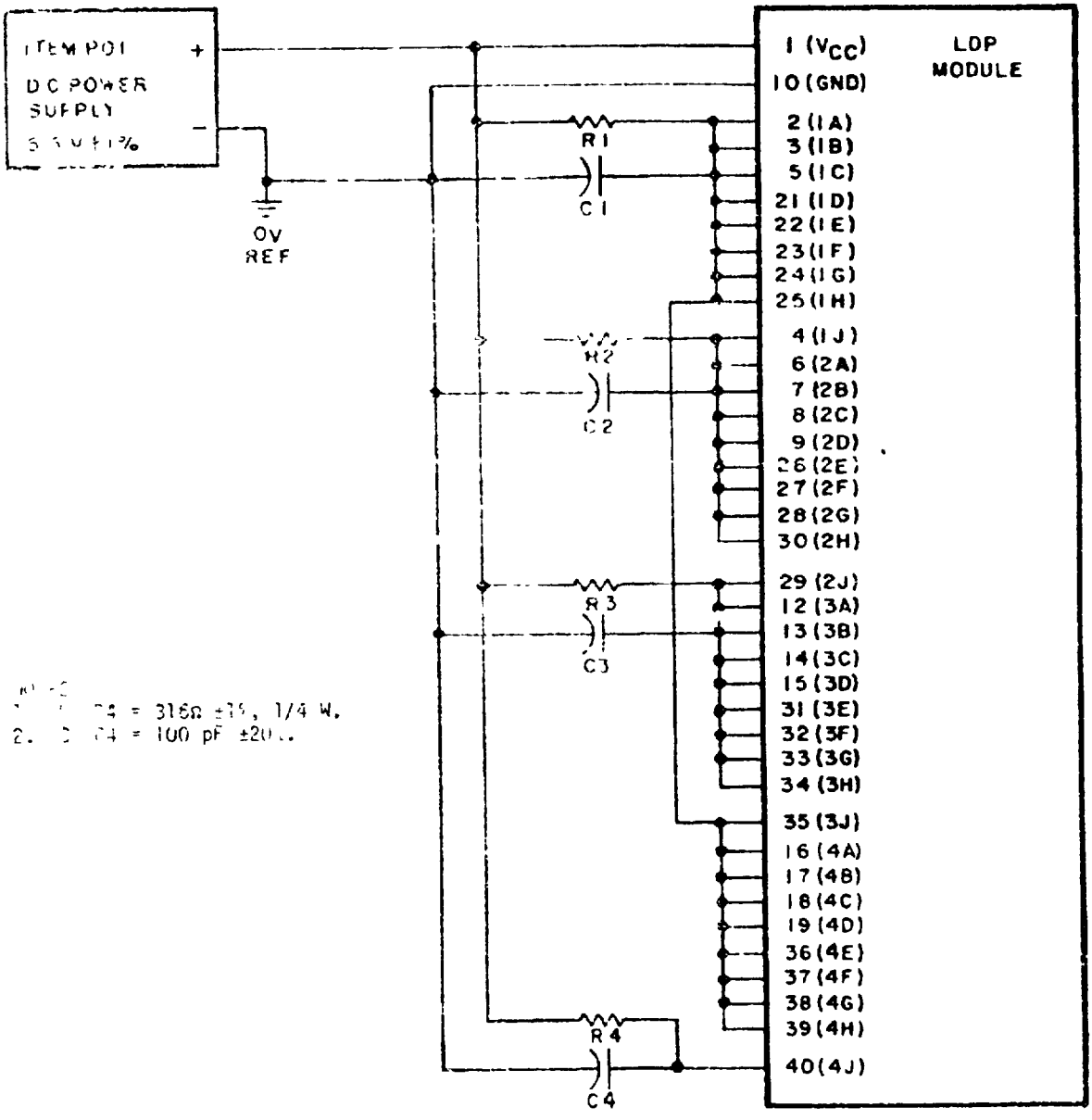
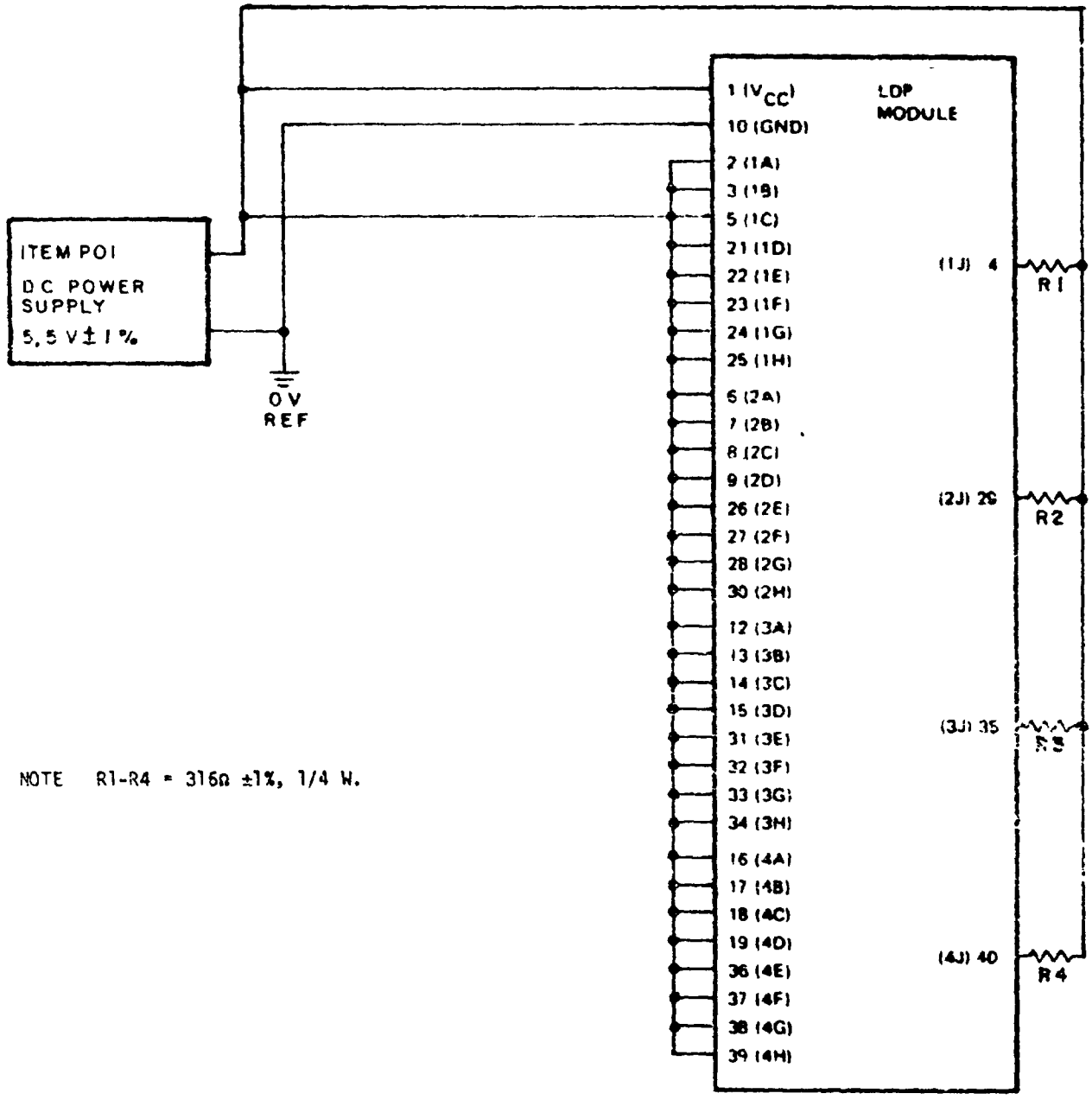


FIGURE 2. Operating temperature and life test circuit.

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NOTE R1-R4 = 316Ω ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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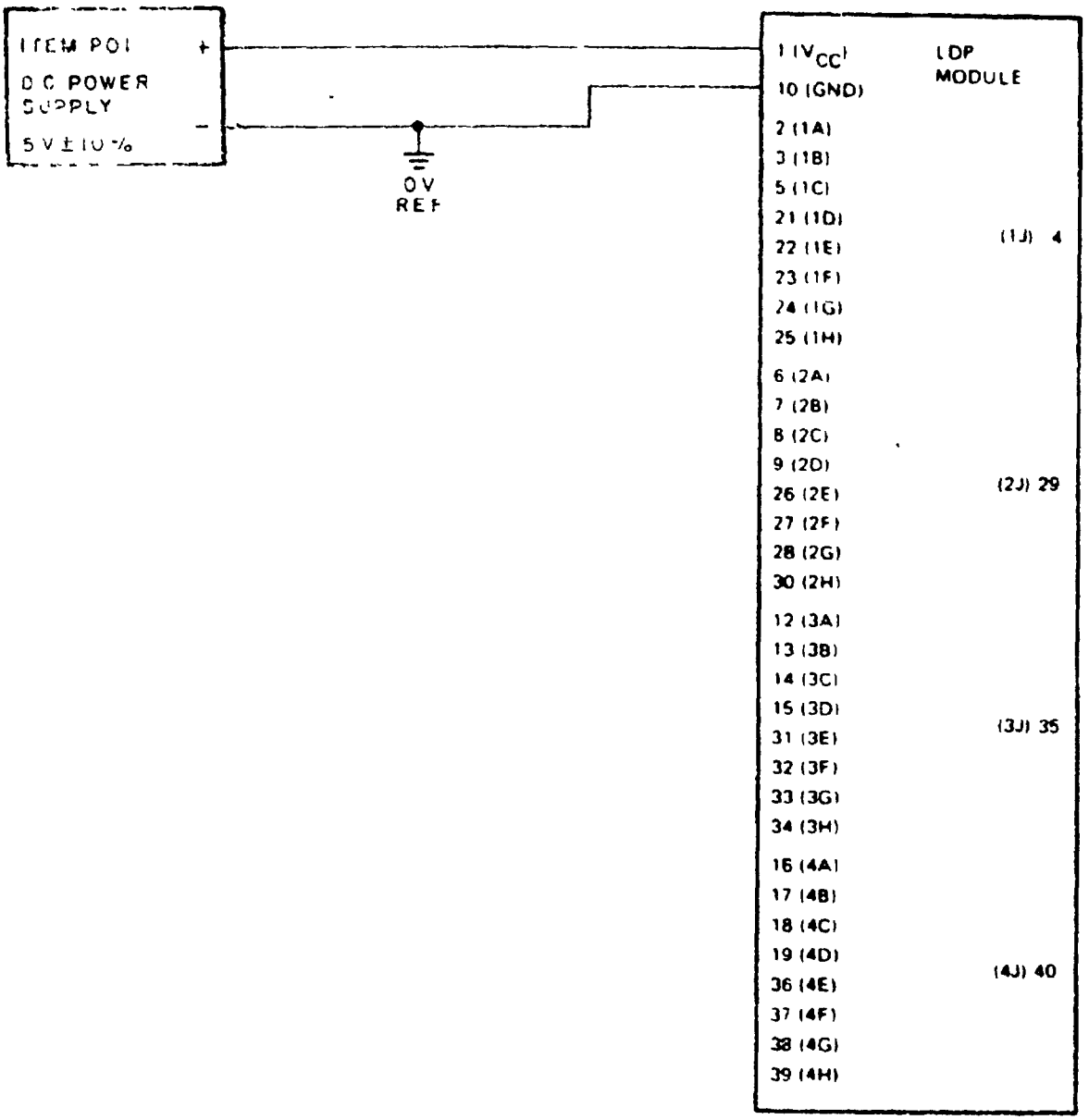


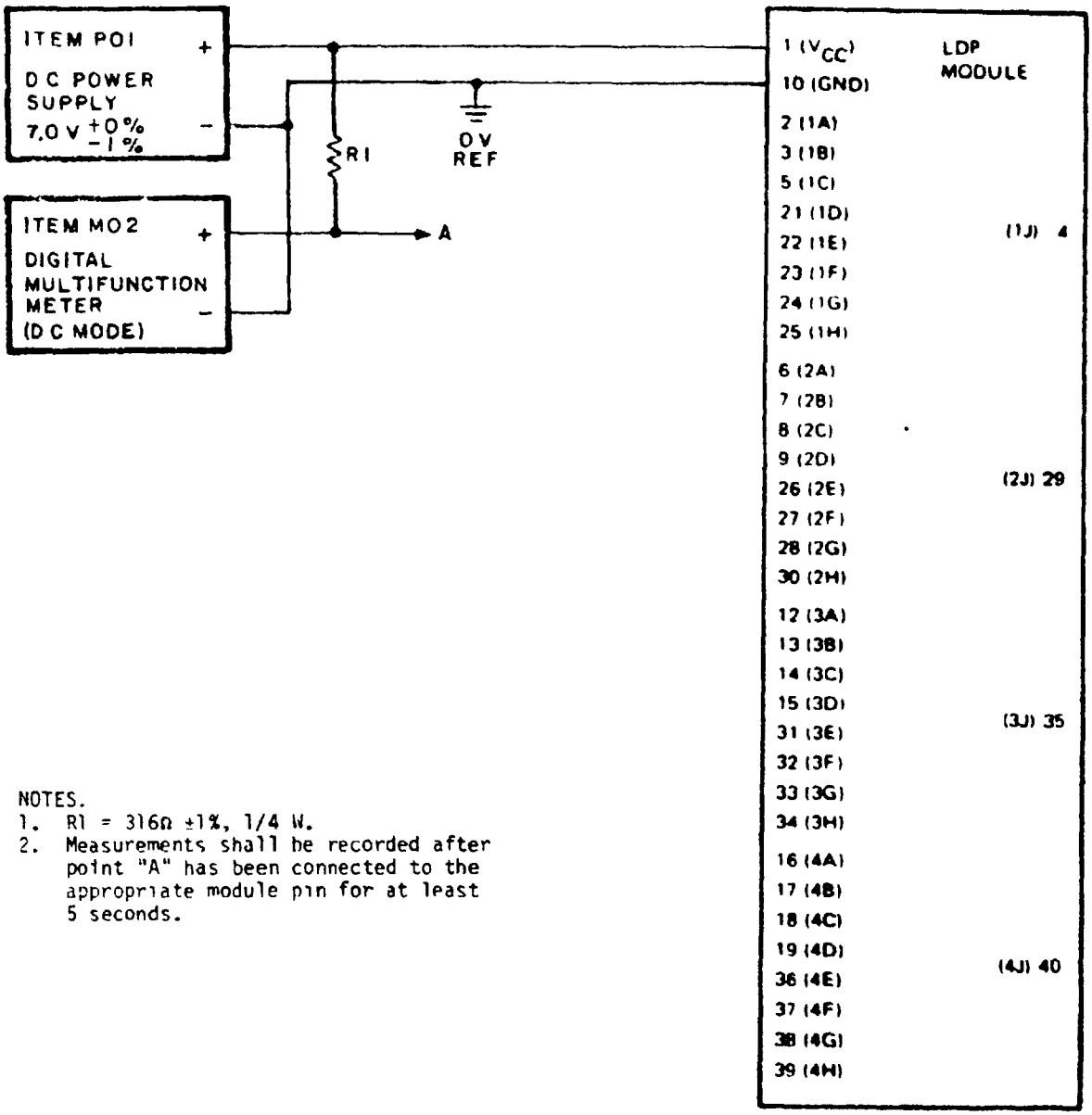
FIGURE 4. Vibration test circuit.

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* TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
A1	1 V_{OUT} (low loaded)	4	10	800	mV	5
A2	2	29	→	→	→	→
A3	3	35	→	→	→	→
A4	4	40	→	→	→	→

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NOTES.
1. R1 = 316Ω ±1%, 1/4 W.
2. Measurements shall be recorded after point "A" has been connected to the appropriate module pin for at least 5 seconds.

FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1A Input breakdown current	2	3,5,21,22,23, 24,25	0	1	mA	6
B2	1B	3	2,5,21,22,23, 24,25				
B3	1C	5	2,3,21,22,23, 24,25				
B4	1D	21	2,3,5,22,23, 24,25				
B5	1E	22	2,3,5,21,23, 24,25				
B6	1F	23	2,3,5,21,22, 24,25				
B7	1G	24	2,3,5,21,22, 23,25				
B8	1H	25	2,3,5,21,22, 23,24				
B9	2A	6	7,8,9,26,27, 28,30				
B10	2B	7	6,8,9,26,27, 28,30				
B11	2C	8	6,7,9,26,27, 28,30				
B12	2D	9	6,7,8,26,27, 28,30				
B13	2E	26	6,7,8,9,27, 28,30				

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TABLE IV. Input breakdown current test - Continued

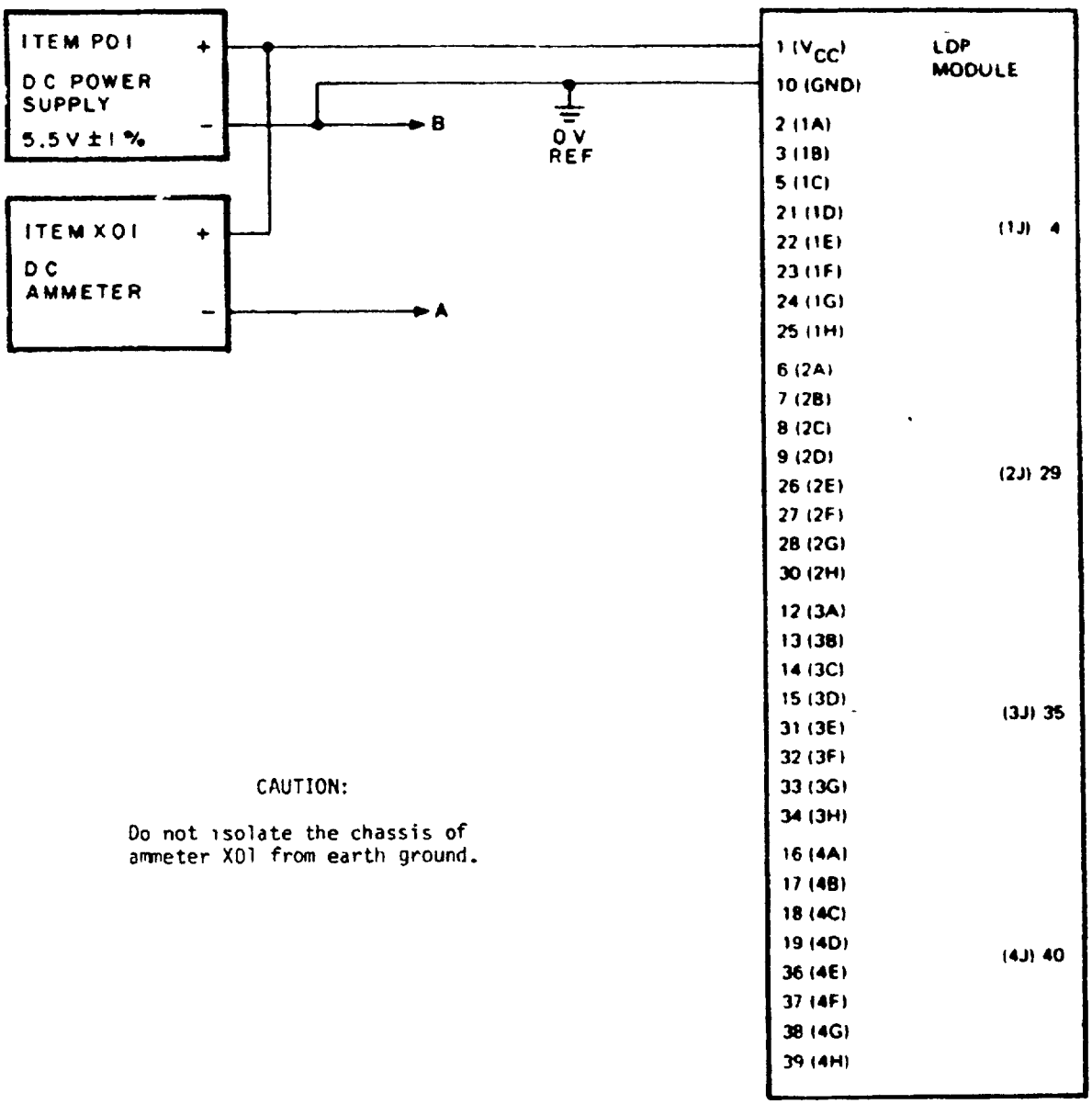
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B14	2F Input breakdown current	27	6,7,8,9,26, 28,30	0	1	mA	6
B15	2G	28	6,7,8,9,26, 27,30				
B16	2H	30	6,7,8,9,26, 27,28				
B17	3A	12	13,14,15,31, 32,33,34				
B18	3B	13	12,14,15,31, 32,33,34				
B19	3C	14	12,13,15,31, 32,33,34				
B20	3D	15	12,13,14,31, 32,33,34				
B21	3E	31	12,13,14,15, 32,33,34				
B22	3F	32	12,13,14,15, 31,33,34				
B23	3G	33	12,13,14,15, 31,32,34				
B24	3H	34	12,13,14,15, 31,32,33				
B25	4A	16	17,18,19,36, 37,38,39				
B26	4B	17	16,18,19,36, 37,38,39				

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TABLE IV. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B27	4C Input breakdown current	18	16,17,19,36, 37,38,39	0	1	mA	6
B28	4D	19	16,17,18,36, 37,38,39				
B29	4E	36	16,17,18,19, 37,38,39				
B30	4F	37	16,17,18,19, 36,38,39				
B31	4G	38	16,17,18,19, 36,37,39				
B32	4H	39	16,17,18,19, 36,37,38				

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

FIGURE 6. Input breakdown current test circuit for table IV.

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* TABLE V. Output voltage (low-level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Output voltage, low level	2,3,5,21,22, 23,24,25	4	0	400	mV	7
C2	↓	6,7,8,9,26, 27,28,30	29	↓	↓	↓	↓
C3		12,13,14,15, 31,32,33,34	35				
C4		16,17,18,19, 36,37,38,39	40				

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TABLE VI. Output voltage (high-level) test.

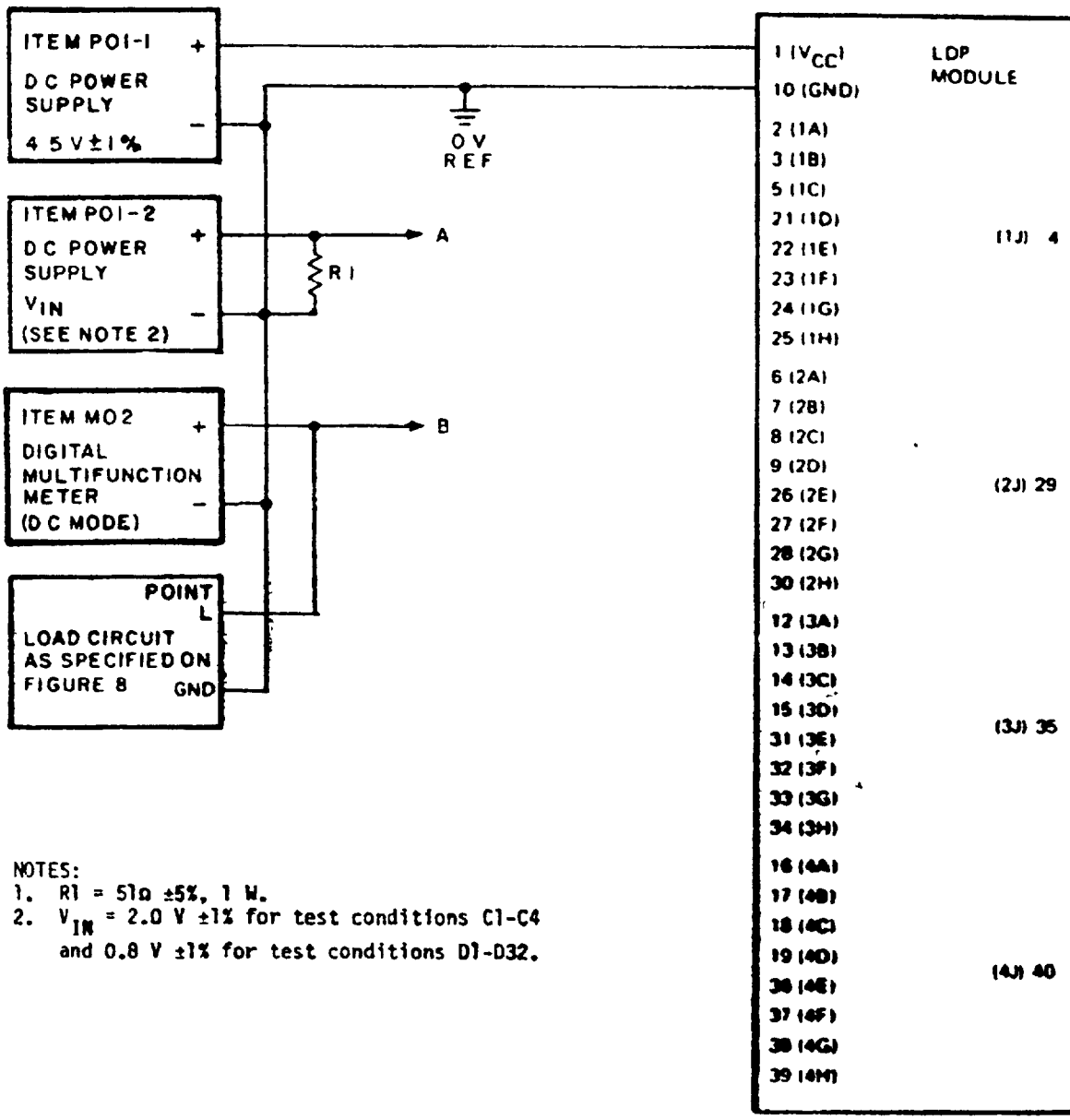
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	Output voltage, high level	2	4	2.4	4.5	V	7
D2		3					
D3		5					
D4		21					
D5		22					
D6		23					
D7		24					
D8		25					
D9		6	29				
D10		7					
D11		8					
D12		9					
D13		26					
D14		27					
D15		28					
D16		30					
D17		12	35				
D18		13	35				
D19		14	35				

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TABLE VI. Output voltage (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D20	Output voltage, high level	15	35	2.4	4.5	V	7
D21		31					
D22		32					
D23		33					
D24		34					
D25		16	40				
D26		17					
D27		18					
D28		19					
D29		36					
D30		37					
D31		38					
D32		39					

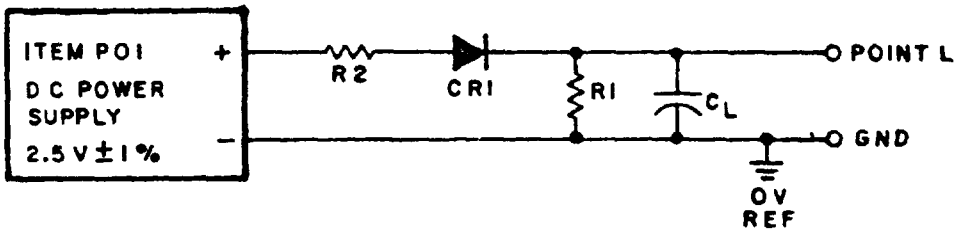
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. V_{IN} = 2.0 V ±1% for test conditions C1-C4 and 0.8 V ±1% for test conditions D1-D32.

* FIGURE 7. Output voltage (low-level and high-level) test circuit for tables V and VI.

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NOTES:
1. C_L = 100 ±5 pF (including probe and parasitic capacitance) for test conditions J, K, L, and M.
2. CR1 = 1N4150 or equivalent.
3. R1 = 6.04 kΩ ±1%, 1/8 W.
4. R2 = select value for current flow of 17.3 mA ±1% from point "L" to ground.
5. C_L is not required for dc electrical tests C1-C4 or D1-D32.
6. The maximum lead length from point "L" to the appropriate module pin shall be 3.00 inches (76.2 mm).
7. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.
8. C_L = 50 ±5 pF for test conditions N and P.

FIGURE 8. Logic load circuit.

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TABLE VII. Input current (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E1	1A Input current, low level	3,5,21, 22,23,24, 25	2	Open	0 40	1 73	mA	9
E2	1B	2,5,21, 22,23,24, 25	3					
E3	1C	2,3,21, 22,23,24, 25	5					
E4	1D	2,3,5,22, 23,24,25	21					
E5	1E	2,3,5,21, 23,24,25	22					
E6	1F	2,3,5,21, 22,24,25	23					
E7	1G	2,3,5,21, 22,23,25	24					
E8	1H	2,3,5,21, 22,23,24	25					
E9	2A	7,8,9,26, 27,28,30	6					
E10	2B	6,8,9,26, 27,28,30	7					
E11	2C	6,7,9,26, 27,28,30	8					
E12	2D	6,7,8,26, 27,28,30	9					

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TABLE VII. Input current (low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E13	2E Input current, low level	6,7,8,9, 27,28,30	26	Open	0.40	1.73	mA	9
E14	2F	6,7,8,9, 26,28,30	27					
E15	2G	6,7,8,9, 26,27,30	28					
E16	2H	6,7,8,9, 26,27,28	30					
E17	3A	13,14,15, 31,32,33, 34	12					
E18	3B	12,14,15, 31,32,33, 34	13					
E19	3C	12,13,15, 31,32,33, 34	14					
E20	3D	12,13,14, 31,32,33, 34	15					
E21	3E	12,13,14, 15,32,33, 34	31					
E22	3F	12,13,14, 15,31,33, 34	32					
E23	3G	12,13,14, 15,31,32, 34	33					

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TABLE VII. Input current (low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E24	3H Input current, low level	12,13,14, 15,31,32, 33	34	Open	0 40	1.73	mA	9
E25	4A	17,18,19, 36,37,38, 39	16					
E26	4B	16,18,19, 36,37,38, 39	17					
E27	4C	16,17,19, 36,37,38, 39	18					
E28	4D	16,17,18, 36,37,38, 39	19					
E29	4E	16,17,18, 19,37,38, 39	36					
E30	4F	16,17,18, 19,36,38, 39	37					
E31	4G	16,17,18, 19,36,37, 39	38					
E32	4H	16,17,18, 19,36,37, 38	39					

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TABLE VIII. Input current (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	1A Input current, high level	2	Open	3,5,21, 22,23,24, 25	0	30	µA	9
F2	1B	3		2,5,21, 22,23,24, 25				
F3	1C	5		2,3,21, 22,23,24, 25				
F4	1D	21		2,3,5,22, 23,24,25				
F5	1E	22		2,3,5,21, 23,24,25				
F6	1F	23		2,3,5,21, 22,24,25				
F7	1G	24		2,3,5,21, 22,23,25				
F8	1H	25		2,3,5,21, 22,23,24				
F9	2A	6		7,8,9,26, 27,28,30				
F10	2B	7		6,8,9,26, 27,28,30				
F11	2C	8		6,7,9,26, 27,28,30				
F12	2D	9		6,7,8,26, 27,28,30				

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TABLE VIII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F13	2E Input current, high level	26	Open	6,7,8,9, 27,28,30	0	30	µA	9
F14	2F	27		6,7,8,9, 26,28,30				
F15	2G	28		6,7,8,9, 26,27,30				
F16	2H	30		6,7,8,9, 26,27,28				
F17	3A	12		13,14,15, 31,32,33, 34				
F18	3B	13		12,14,15, 31,32,33, 34				
F19	3C	14		12,13,15, 31,32,33, 34				
F20	3D	15		12,13,14, 31,32,33, 34				
F21	3E	31		12,13,14, 15,32,33, 34				
F22	3F	32		12,13,14, 15,31,33, 34				

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TABLE VIII. Input current (high-level) test - Continued.

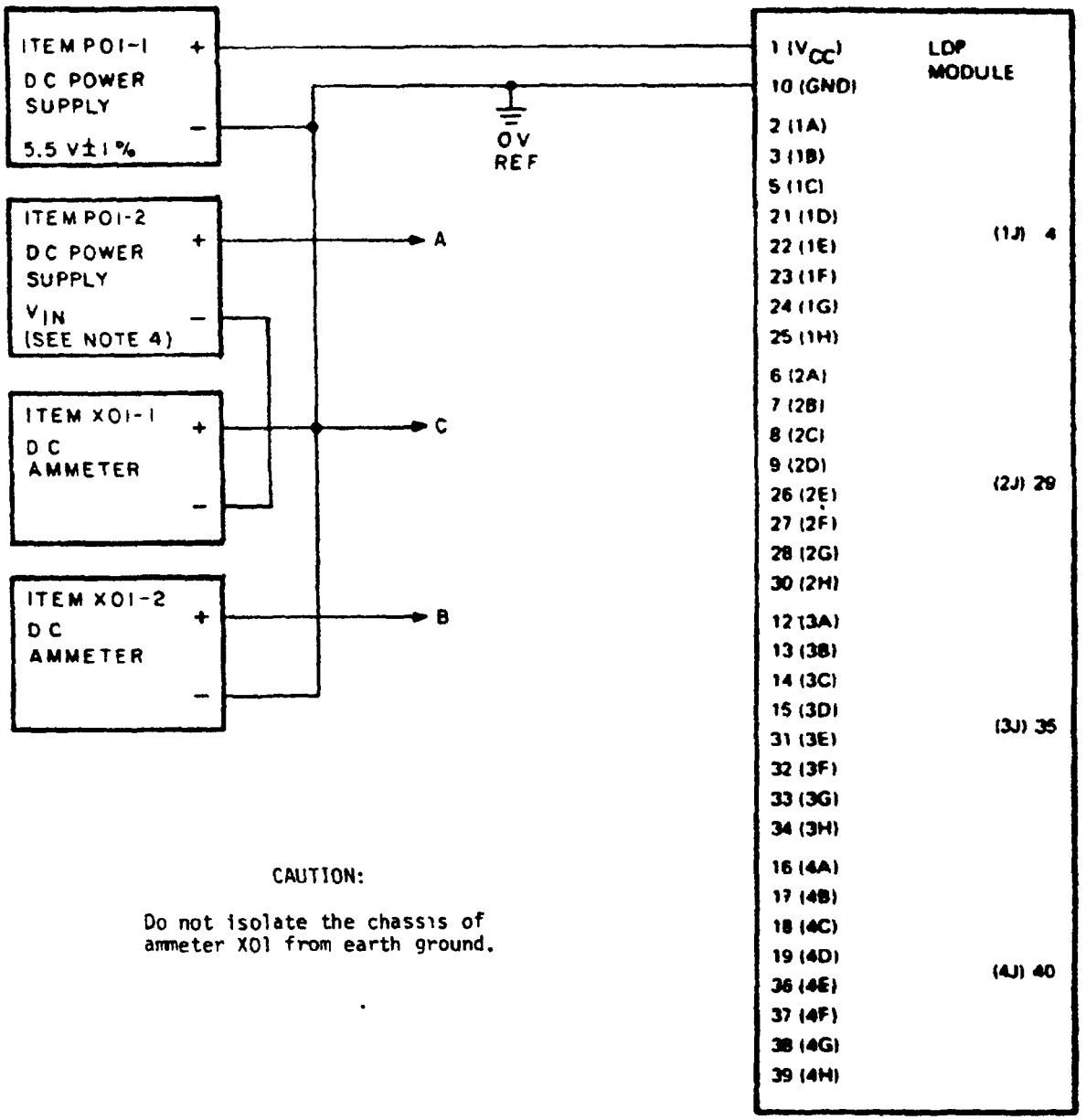
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F23	3G Input current, high level	33	Open	12,13,14, 15,31,32, 34	0	30	µA	9
F24	3H	34		12,13,14, 15,31,32, 33				
F25	4A	16		17,18,19, 36,37,38, 39				
F26	4B	17		16,18,19, 36,37,38, 39				
F27	4C	18		16,17,19, 36,37,38, 39				
F28	4D	19		16,17,18, 36,37,38, 39				
F29	4E	36		16,17,18, 19,37,38, 39				
F30	4F	37		16,17,18, 19,36,38, 39				
F31	4G	38		16,17,18, 19,36,37, 39				
F32	4H	39		16,17,18, 19,36,37, 38				

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TABLE IX. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	Output short circuit current	Open	4	2	20	55	mA	9
G2	↓	↓	29	6	↓	↓	↓	↓
G3			35	12				
G4			40	16				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required. Use item X01-2 for tables VII and IX; use item X01-1 for table VIII.
2. With point A disconnected, item X01-1 should indicate zero current on the 1 µA range.
3. When performing output short circuit tests, only one output shall be shorted at a time and the measurement taken within 5 seconds.
4. V_{IN} = 4.5 V ±1% for test conditions E1-E32 and 2.4 V ±1% for test conditions F1-F32.

FIGURE 9. Input current (low-level and high-level) and output short circuit current test circuit for tables VII, VIII, and IX.

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TABLE X. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	Power supply current, low level	Open	Open	1	24	mA	10
H2	Power supply current, high level	B	A	1	8	mA	10

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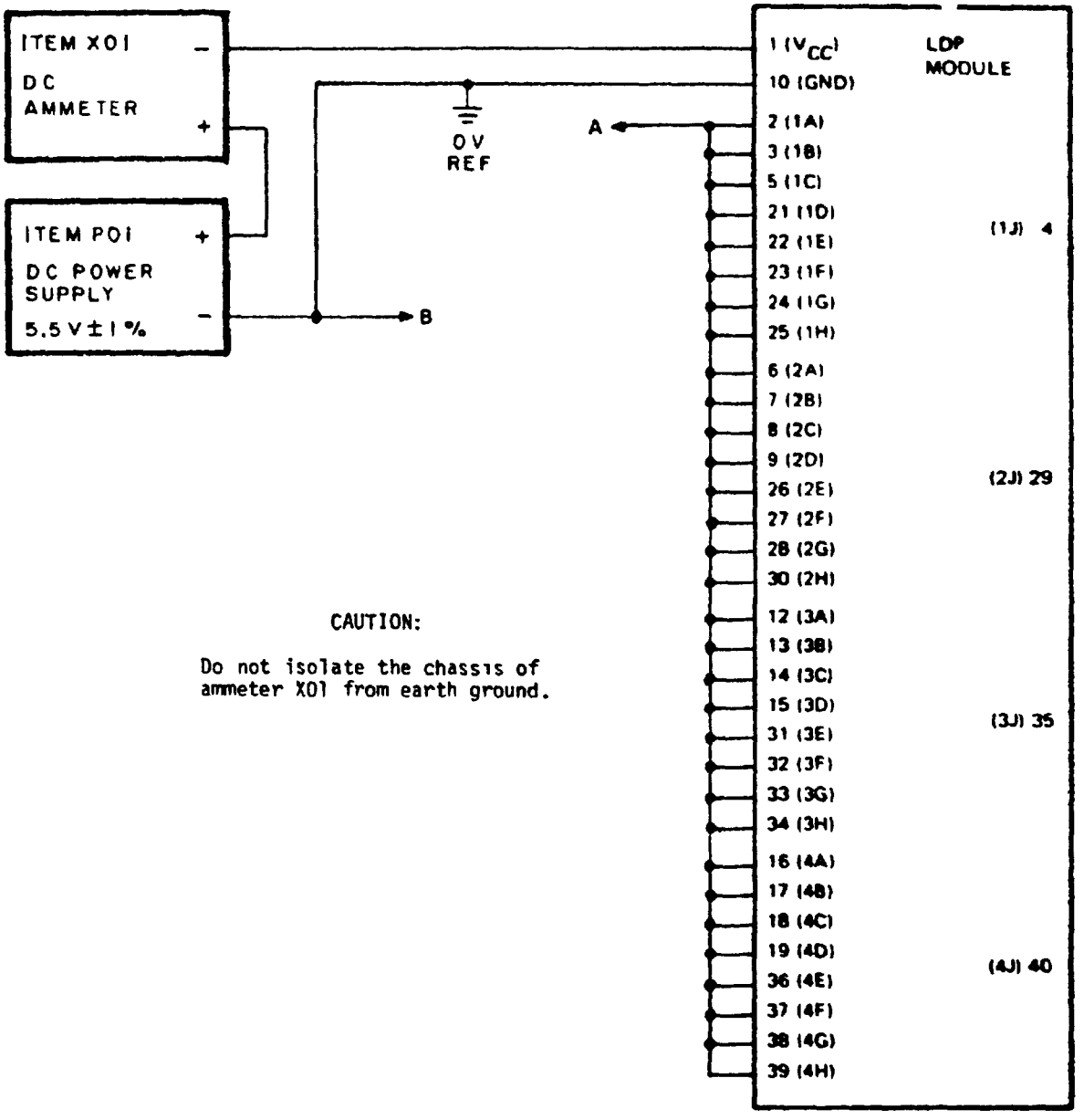


FIGURE 10. Power supply current (low-level and high-level) test circuit for table X.

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TABLE XI. Transition time (t_{TLH} and t_{THL}) test.

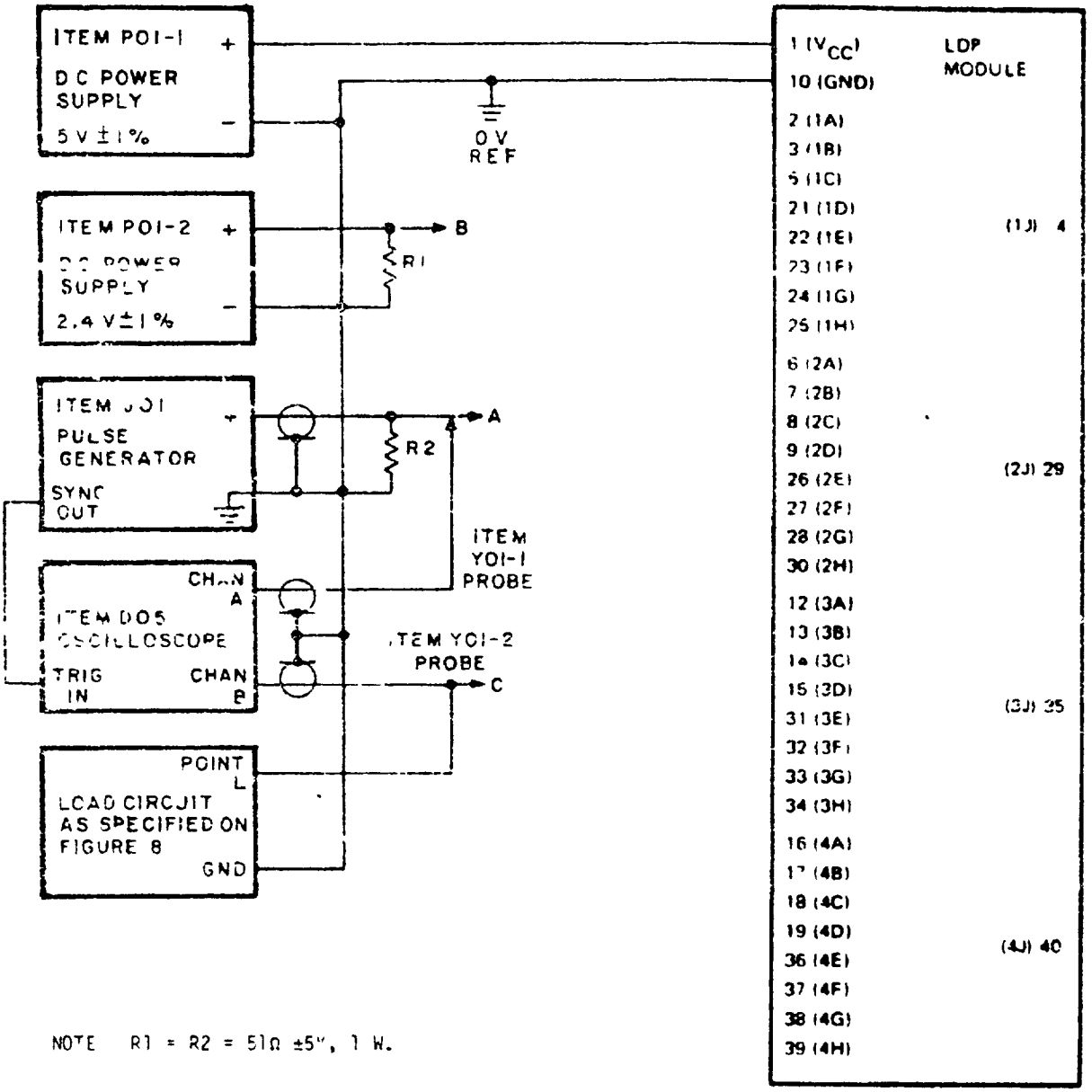
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	Transition time (t_{TLH})	2	3,5,21,22, 23,24,25	4	4	45	ns	11,12
J2		6	7,8,9,26, 27,28,30	29				
J3		12	13,14,15, 31,32,33, 34	35				
J4		16	17,18,19, 36,37,38, 39	40				
K1	Transition time (t_{THL})	2	3,5,21,22, 23,24,25	4		20		
K2		6	7,8,9,26, 27,28,30	29				
K3		12	13,14,15, 31,32,33, 34	35				
K4		16	17,18,19, 36,37,38, 39	40				

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TABLE XII. Propagation delay time (t_{PHL} and t_{PLH}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	Propagation delay time (t_{PHL})	2	3,5,21,22, 23,24,25	4	4	25	ns	11,12
L2		6	7,8,9,26, 27,28,30	29				
L3		12	13,14,15, 31,32,33, 34	35				
L4		16	17,18,19, 36,37,38, 39	40				
M1	Propagation delay time (t_{PLH})	2	3,5,21,22, 23,24,25	4		28		
M2		6	7,8,9,26, 27,28,30	29				
M3		12	13,14,15, 31,32,33, 34	35				
M4		16	17,18,19, 36,37,38, 39	40				

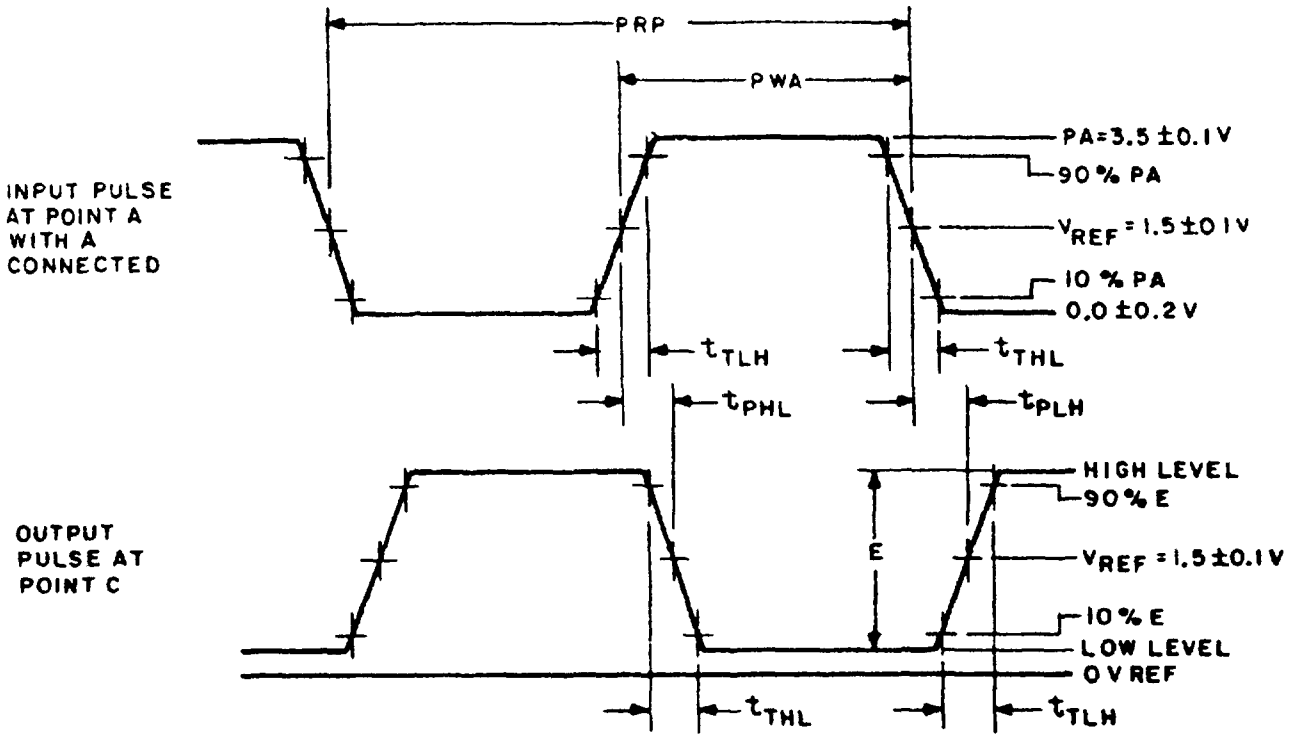
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NOTE R1 = R2 = 51Ω ±5%, 1 W.

* FIGURE 11. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit for tables XI and XII.

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NOTES:
1. Input pulse parameters:
   PWA = 100 ns ±10%.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   PRP = 200 ns ±10%.
2. All voltage levels are referenced to ground.
3. Verify that the outputs have a high level $\geq$ 2.4 V and a low level $\leq$ 0.4 V before making measurements.

* FIGURE 12. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for tables XI and XII.

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TABLE XIII Output crosstalk (low-level) test.

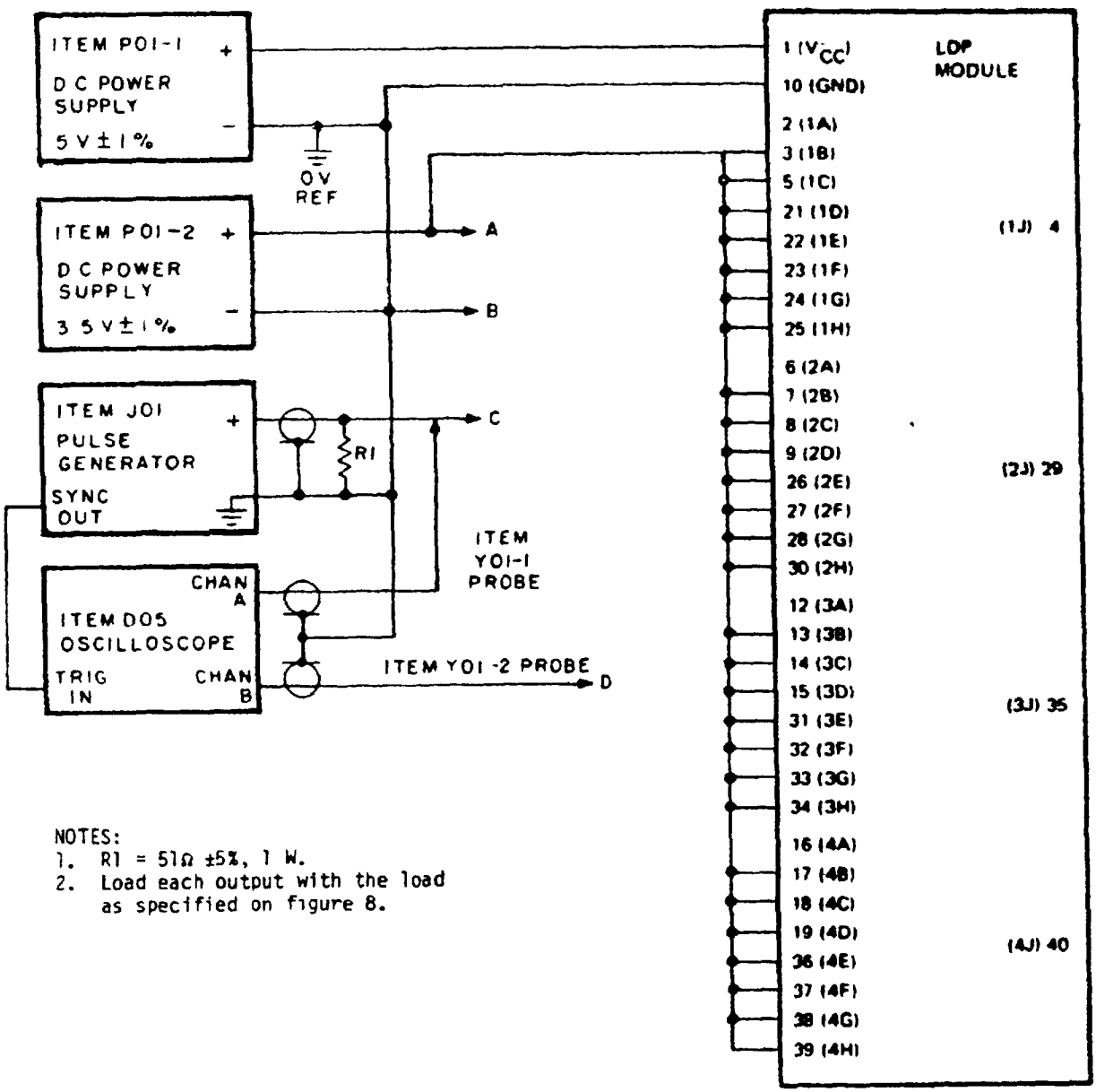
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	Output crosstalk, low level	2	Open	6,12,16	4	0	400	mV	13,14
N2	↓	6	↓	2,12,16	29	↓	↓	↓	↓
N3	↓	12	↓	2,6,16	35	↓	↓	↓	↓
N4	↓	16	↓	2,6,12	40	↓	↓	↓	↓

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TABLE XIV Output crosstalk (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	Output crosstalk, high level	Open	2	6,12,16	4	0	400	mV	13,14
P2	↓	↓	6	2,12,16	29	↓	↓	↓	↓
P3			12	2,6,16	35				
P4			16	2,6,12	40				

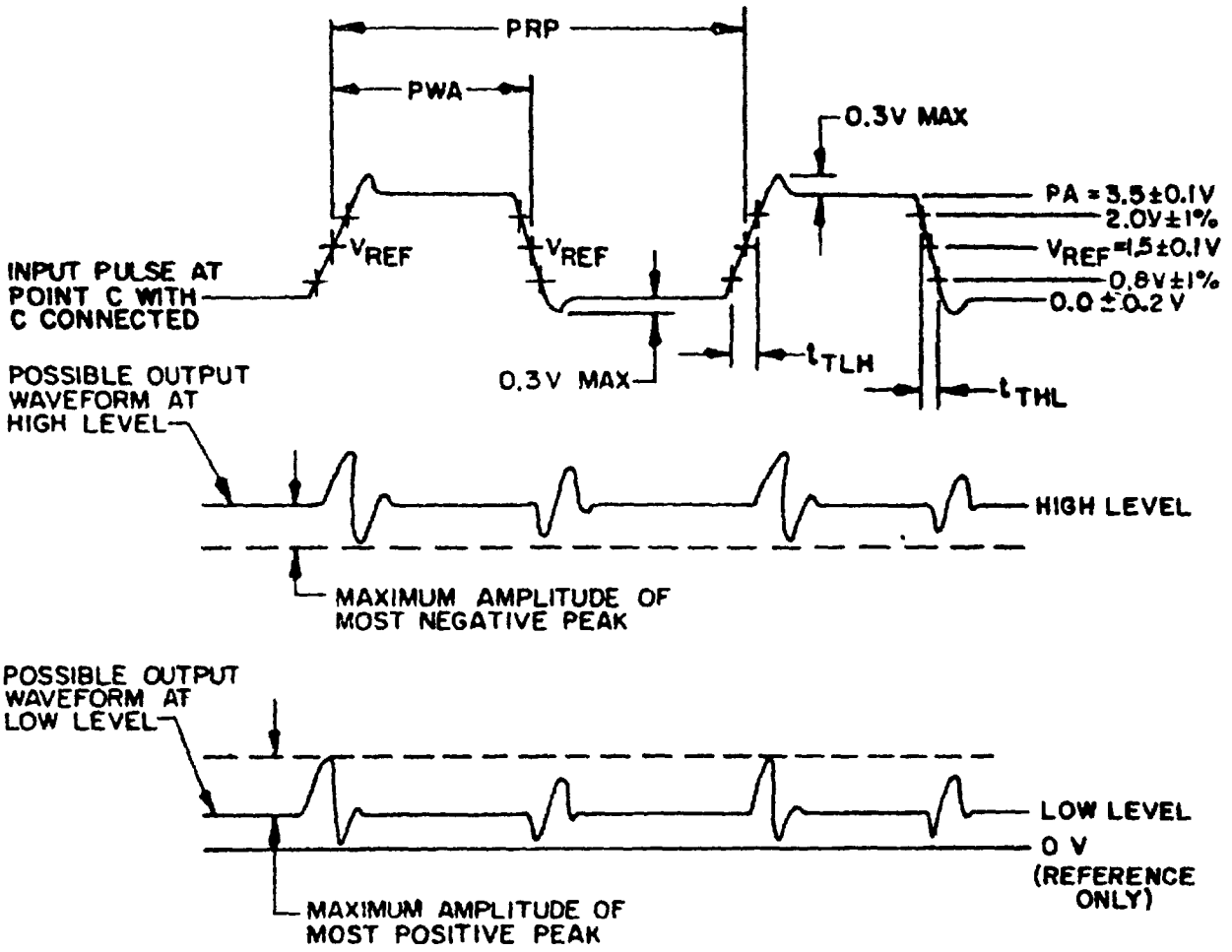
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. Load each output with the load as specified on figure 8.

FIGURE 13. Output crosstalk (low-level and high-level) test circuit for tables XIII and XIV.

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NOTES:
1. Input pulse parameters:
   $t_{THL} = t_{TLH}$ = 3.5 ±1.5 ns (verified at all input pins).
   PWA = 200 ±20 ns.
   PRP = 1,000 ±100 ns.
2. The lead length of electrical conductors shall be less than two inches (50.80 mm) between the resistor R1 and the input pins; probes YO1-1 and YO1-2 and the input and output pins; load circuits and the output pins; and ground leads of the probes YO1-1 and YO1-2 and the test fixture and module ground pins.

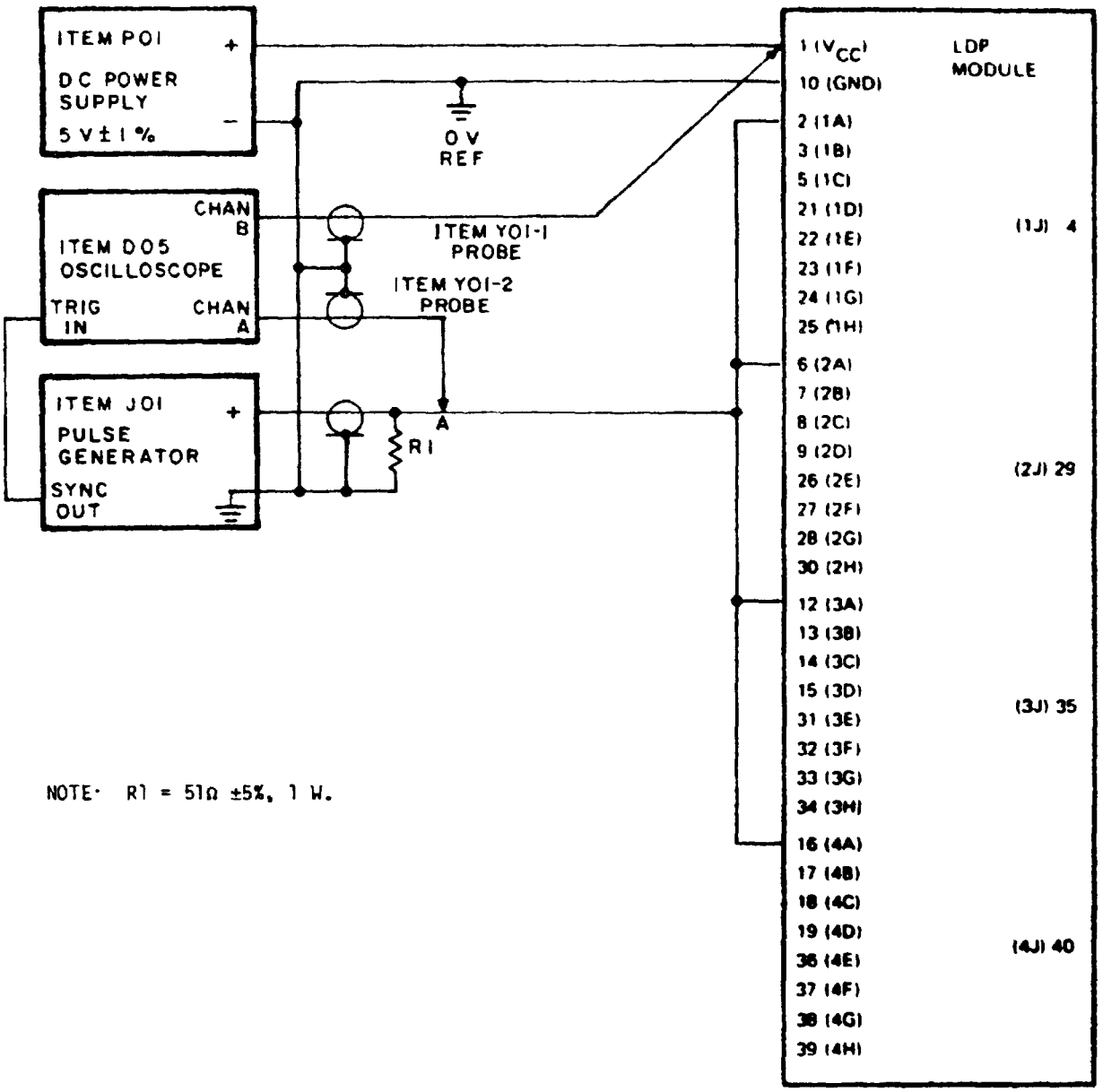
* FIGURE 14. Output crosstalk (low-level and high-level) test circuit waveforms for tables XIII and XIV.

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TABLE XV. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
R	Power supply transient amplitude	0	400	mV	15,16

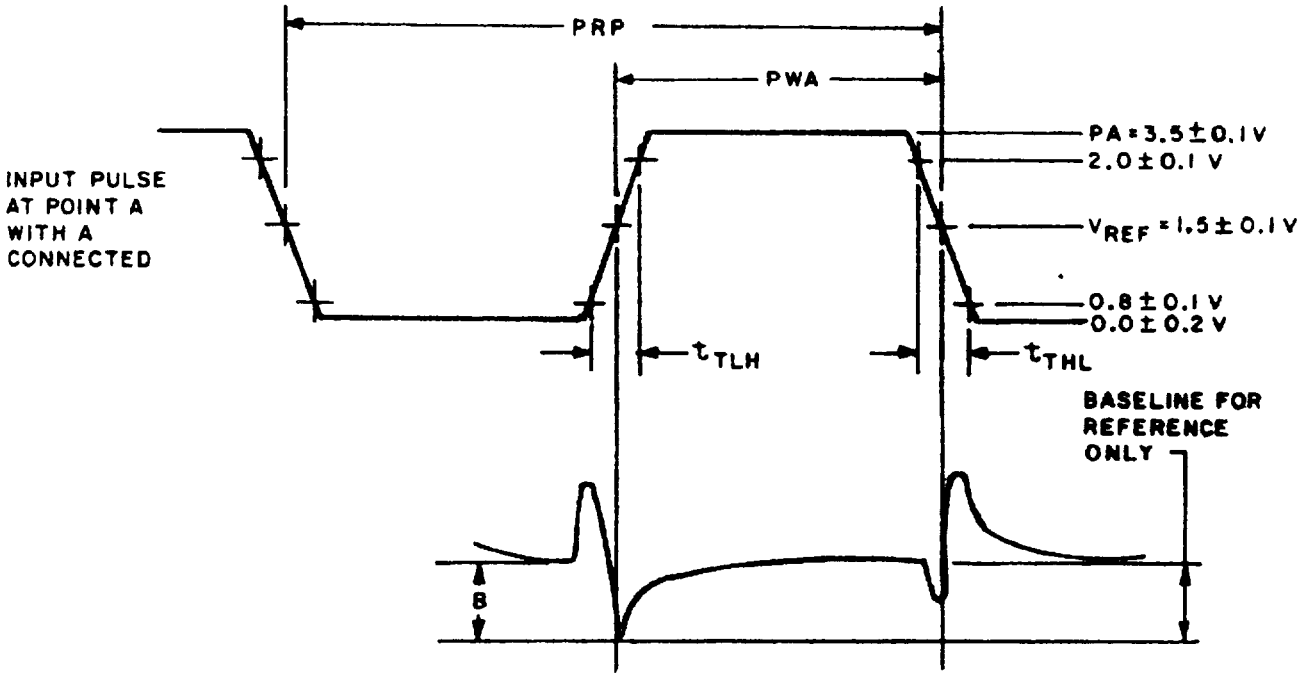
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NOTE: R1 = 51Ω ±5%, 1 W.

* FIGURE 15. Power supply transient amplitude test circuit for table XV.

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NOTES:
1. Input pulse parameters:
   PWA = 3 µs ±10%.
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
   PRP = 10 µs ±1%.
2. B = measurement from the baseline to the most negative excursion of the waveform.

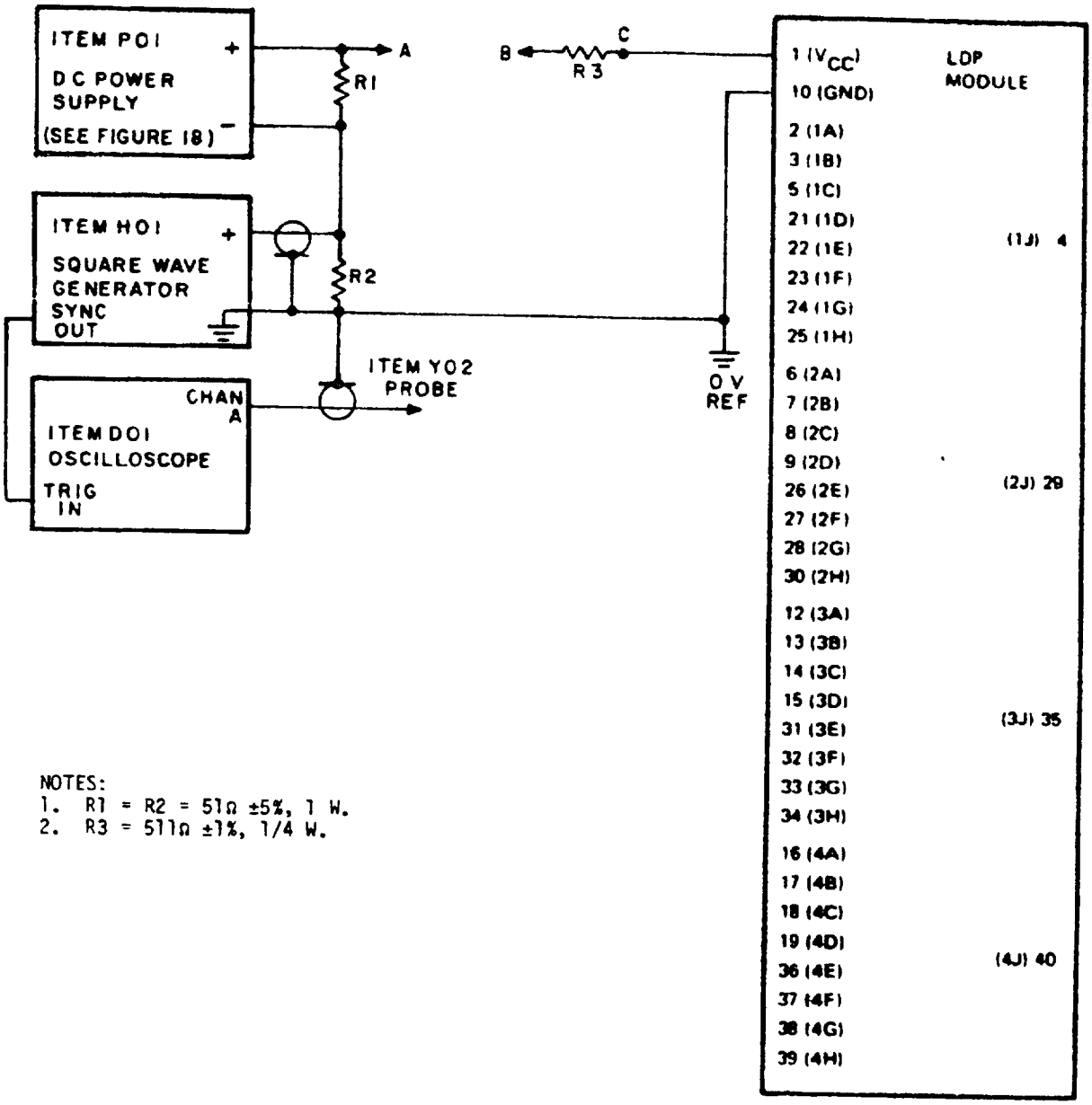
* FIGURE 16. Power supply transient amplitude test circuit waveforms for table XV.

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TABLE XVI. Power supply transient rise time test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
S	Power supply transient rise time	1	2	ms	17,18

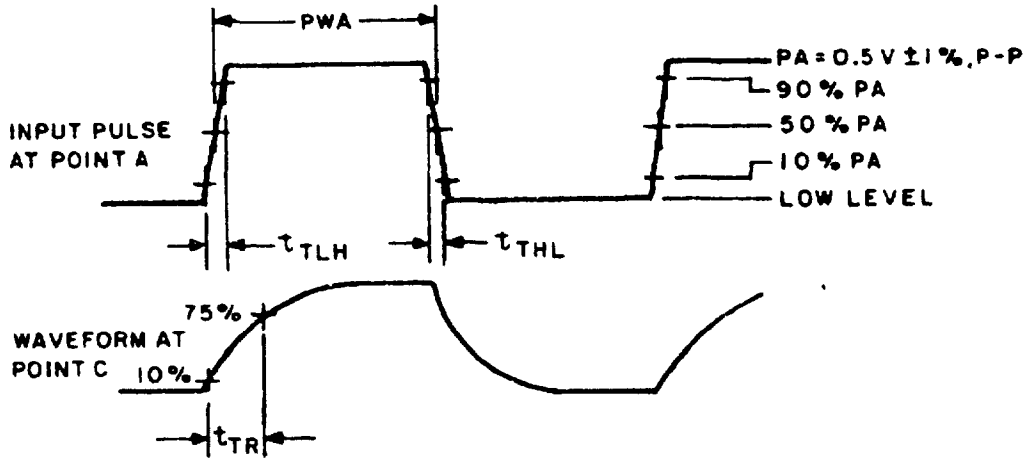
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. R3 = 511Ω ±1%, 1/4 W.

* FIGURE 17. Power supply transient rise time test circuit for table XVI.

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NOTES:
1. Input pulse parameters:
   PWA = 10 ms ±10%.
   PRR = 50 Hz (reference).
   t_{TLH} and $t_{THL} \leq$ 100 ns (reference).
2. With point A open, adjust input as shown.
3. With point A open, adjust item P01 such that the low level of the input is at 0 ±0.01 V.
4. Connect A to B; recheck the low level and pulse parameters at point A.
5. Measure transient rise time t_{TR} at point C.

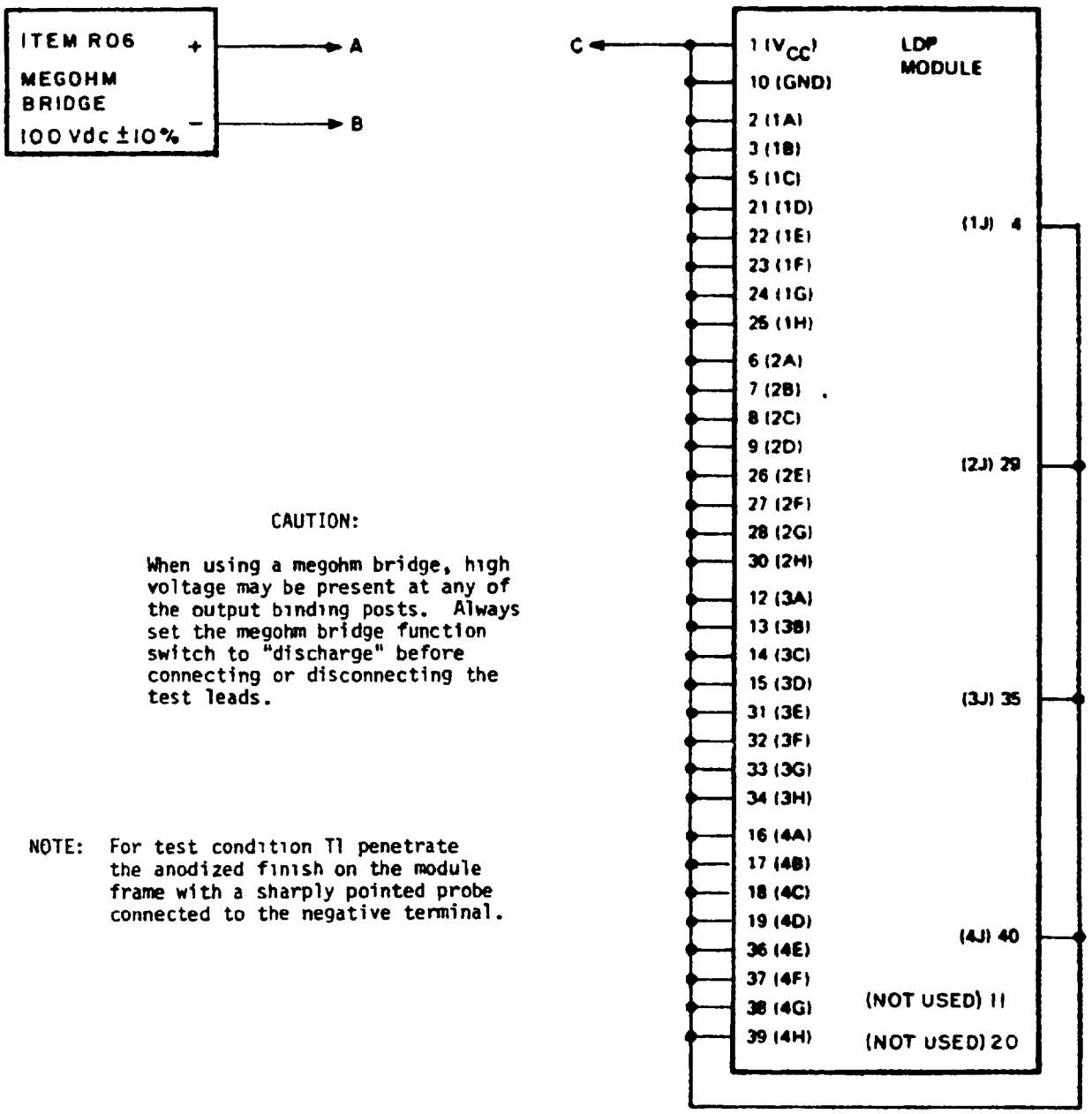
* FIGURE 18. Power supply transient rise time test circuit waveforms for table XVI.

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TABLE XVII. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T1	Isolation	C,11,20	Frame	10	∞	MΩ	19
T2	Unused pin isolation	C	11,20	10	∞	MΩ	19
T3	Unused pin isolation	11	20	10	∞	MΩ	19

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CAUTION:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition T1 penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 19. Isolation test circuit for table XVII.

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TABLE XVIII. Input capacitance test.

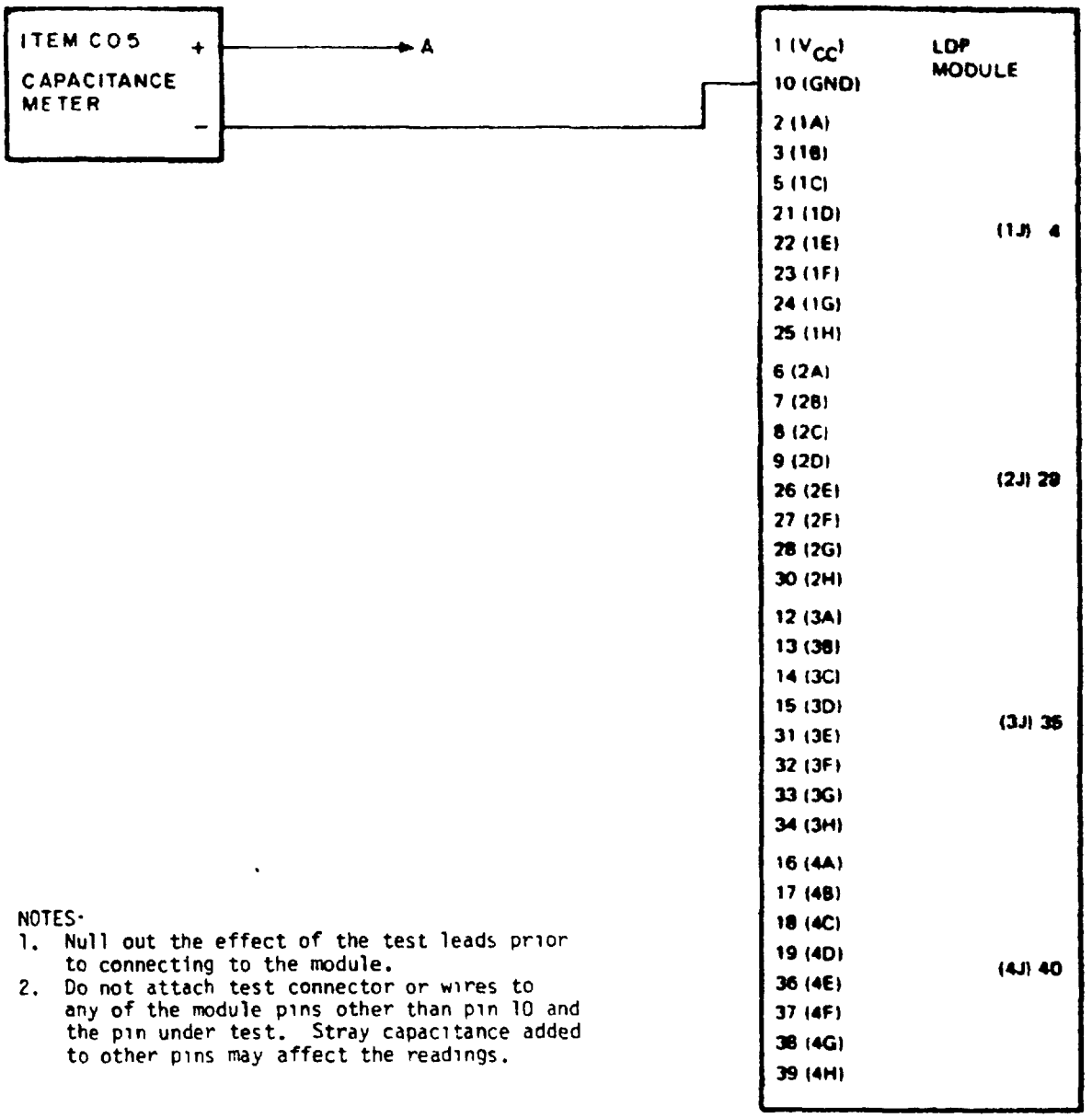
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
U1	1A Input capacitance	2	2	15	pF	20
U2	1B	3				
U3	1C	5				
U4	1D	21				
U5	1E	22				
U6	1F	23				
U7	1G	24				
U8	1H	25				
U9	2A	6				
U10	2B	7				
U11	2C	8				
U12	2D	9				
U13	2E	26				
U14	2F	27				
U15	2G	28				
U16	2H	29				
U17	3A	12				
U18	3B	13				

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TABLE XVIII. Input capacitance test - Continued.

Test condition	Parameter		Connect	Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
U19	3C	Input capacitance	14	2	15	pF	20
U20	3D		15				
U21	3E		31				
U22	3F		32				
U23	3G		33				
U24	3H		34				
U25	4A		16				
U26	4B		17				
U27	4C		18				
U28	4D		19				
U29	4E		36				
U30	4F		37				
U31	4G		38				
U32	4H		39				

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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

FIGURE 20. Input capacitance test circuit for table XVIII.

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TABLE XIX. Input clamp diode voltage test.

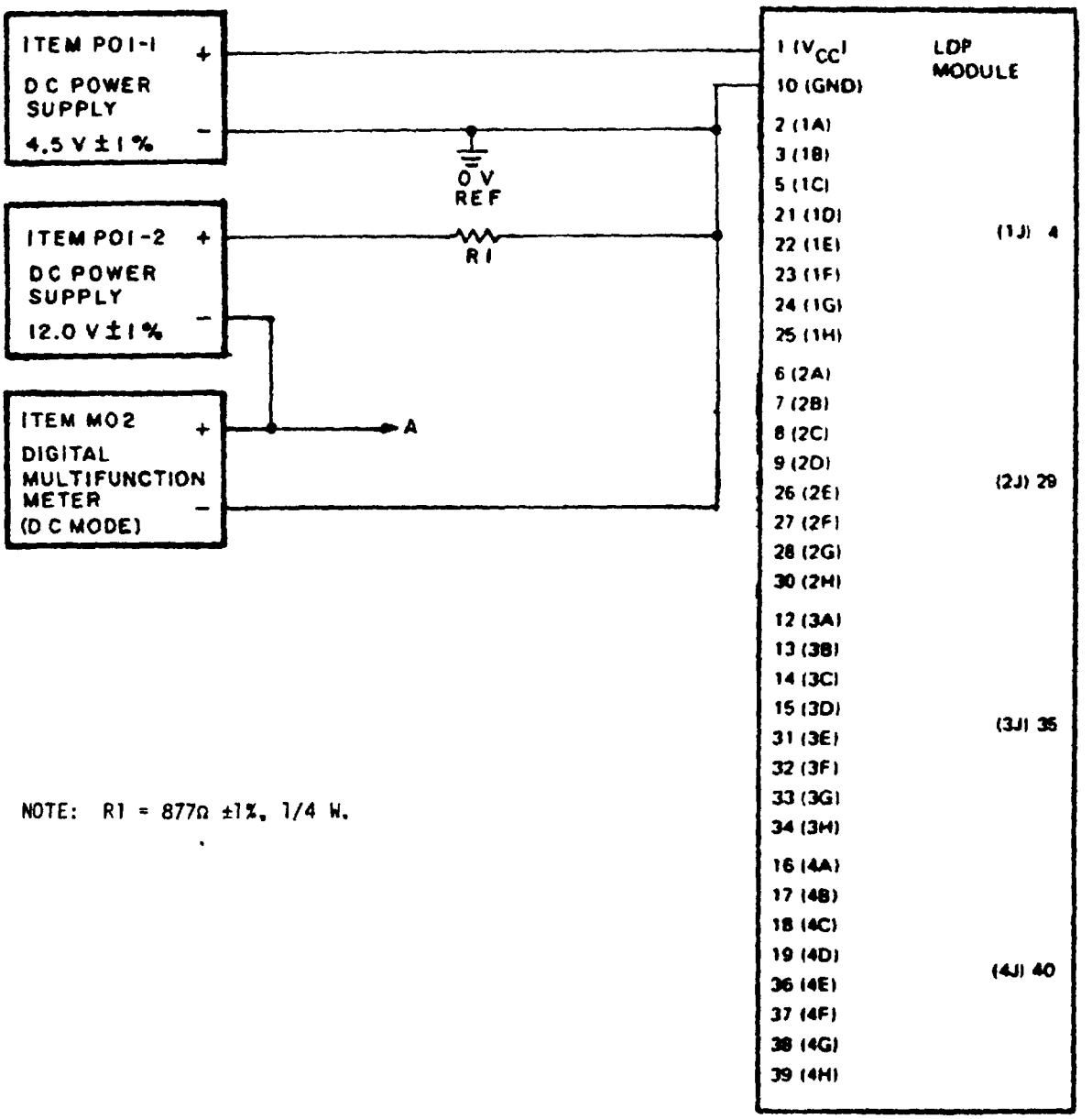
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
V1	Input clamp diode voltage	2	-0.4	-1.5	V	21
V2		3				
V3		5				
V4		21				
V5		22				
V6		23				
V7		24				
V8		25				
V9		6				
V10		7				
V11		8				
V12		9				
V13		26				
V14		27				
V15		28				
V16		30				
V17		12				
V18		13				
V19		14				

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TABLE XIX. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
V20	Input clamp diode voltage	15	-0.4	-1.5	V	21
V21	↓	31	↓	↓	↓	↓
V22		32				
V23		33				
V24		34				
V25		16				
V26		17				
V27		18				
V28		19				
V29		36				
V30		37				
V31		38				
V32		39				

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NOTE: R1 = 877Ω ±1%, 1/4 W.

FIGURE 21. Input clamp diode test circuit for table XIX.

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MIL-M-28787/9B
AMENDMENT 2
31 March 1993
SUPERSEDING
AMENDMENT 1
8 March 1984

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE LDP

This amendment forms a part of Military Specification MIL-M-28787/9B(EC), dated 22 June 1982, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 4

*TABLE I, Test conditions G1-G4, Test type column: Delete "X" and substitute "S".

PAGE 5

*TABLE I, Test conditions T2 and T3, Test type column: Delete "Q" and substitute "S".

*TABLE I, Test conditions V1-V32, Test type column: Delete "Q" and substitute "S".

PAGE 6

*4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through V shall be performed in any order with the following exceptions. Test conditions A, B, G, T, and V are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, G, T, or V are part of a sequence of tests, the order of testing shall be as follows: T, A, B, G, and V where applicable, followed by the remaining tests in the sequence."

*4.5.1, line 5: Delete "C through M" and substitute "C through F, H through M".

*4.5.3, lines 3 and 4: Delete "C through T and V" and substitute "C through F, and H through S".

*4.5.3, line 6: Delete "C through V" and substitute "C through F, H through S, and U".

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